

EC331 Project update - Prof Alan Manning

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Specification

OLS:

$$Y_{dt} = \alpha + \beta Protection_{dt} + \gamma_d + \lambda_t + \epsilon_{dt}$$

where $Protection_{dt} = \sum_j w_{dj} \cdot Tariff_{jt}$ by identity; the net exposure to tariffs in each region

SSIV regression - reduced form specification:

$$Y_{dt} = \alpha + \beta \sum_j w_{dj,1991} \cdot Tariff_{j1991,1997} + \gamma_d + \lambda_t + \epsilon_{dt}$$

Clustered standard errors at the composite region level, but will compute exposure-robust SEs (ssaggregate package, following Borusyak, Hull and Jaravel)

Descriptive figures and preliminary results

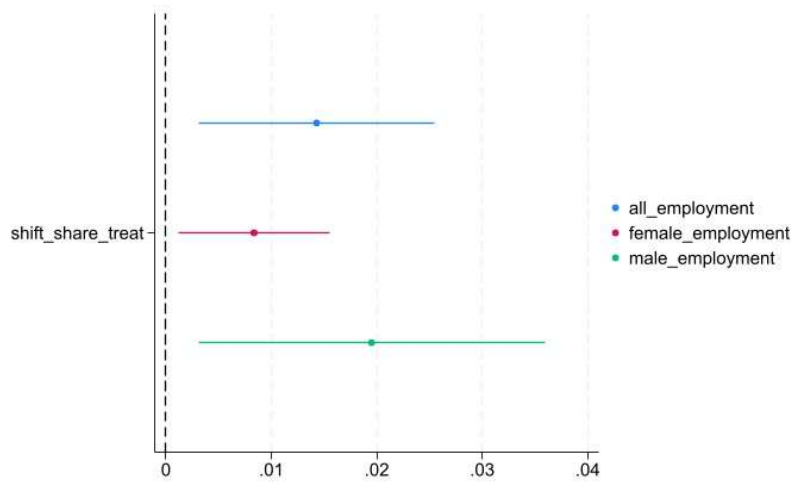


Figure 1: Overall employment coefficient plot from SSIV regression

Mean values for: all employment: 0.403, male employment: 0.537, female employment: 0.173. Sample: 747 observations

VARIABLES	Rural Employment			Urban Employment		
	(1)	(2)	(3)	(4)	(5)	(6)
	Overall	Female	Male	Overall	Female	Male
Shift-share treatment	0.028	0.011	0.041	-0.001*	0.001	-0.002**
	(0.020)	(0.012)	(0.032)	(0.000)	(0.001)	(0.001)
Clustered standard errors, at level of region, in parentheses						
*** p<0.01, ** p<0.05, * p<0.1						

Figure 2: Regression table from SSIV regression on rural and urban parts of a region
S.d of employment for urban regions is 0.0419- could be what is driving these results

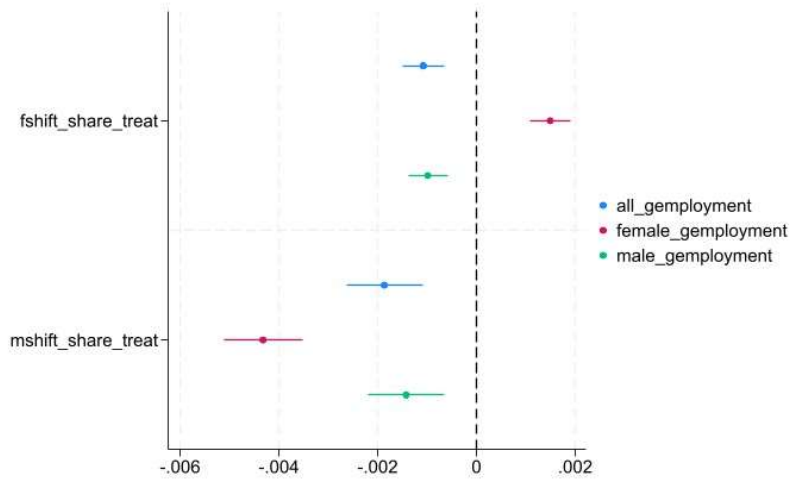


Figure 3: Coefficient plots from SSIV regression, using gender-specific components of region's trade exposure
(Treatment and instrument: summation of gendered industry shares in region)

Thoughts and concerns

- **Employment as an outcome:** Insufficient variation within districts across time, particularly for urban parts of a district.
 - Accounting for informal employment in India: incorporating marginal workers (have not worked for major part of reference period - less than 6 months) into analysis - add to industry shares and total employment figures. (Would this have similar variation to employment within districts across time?)
 - Gendered employment outcomes: looking at gender-specific exposure to trade on female employment (s.d of female employment within districts across time: 0.029, mean overall: 0.175)
- **Additional outcomes:**
 - Fertility and women's age at marriage (using census data).
 - Total road length and number of markets in a district - can test connectivity hypotheses - greater exposure to trade leading to increased market access/infrastructure?
 - Household assets (census), including number of households using banking services. Potentially would be more informative for rural regions (if many urban households have these assets regardless).
- **Accounting for non-traded agricultural goods:** Better capturing a region's exposure to trade, and particularly with crops - confounders (in the literature, poorer regions tend to have greater shares of non-traded crops - mean reversion with outcomes like employment). Identified crops which were cannalised until 1991.
 - Controlling for % of gross cropped area cultivating non-traded crops - variation would be netted out
 - Using this above percentage to discount share of workers as agricultural labourers/cultivators - but depending on cultivation processes, may be inaccurate
 - Using ICRISAT - would restrict sample to districts which have unchanged boundaries from 1966 (composite region analysis introduces some difficulty).